

Use case	Search Train	
Release (id)	UC-01, v1.0, 26 April 2026	
Context of Use	A user (anonymous or registered) plans a trip and searches for available train connections, optionally using map selection and intelligent filters.	
Scope	Allow the user to find trains by specifying departure/arrival, date, time, class, and filters; visualise results and trip details.	
Level	Primary task	
Primary actor	General User (unregistered or registered)	
StakeHolder & interest	StakeHolder	Interest
	General User	Wants fast, accurate search results with delay prediction and map interaction
	Railway administrator	Needs correct train data and usage statistics from search behaviour
Preconditions	→ The system is connected to the Train Database and Prediction System → The Mapping System API is available (for FR 11)	
Minimal Guarantees	→ The system returns a list of trains even if the Prediction System is temporarily unreachable → No incorrect train data is displayed	
Success Guarantees	→ The user sees a list of trains matching the search criteria → Delay predictions and trip details are shown correctly → Map-based station selection works if used	
Trigger	The user accesses the search page and clicks “Search” after entering departure/arrival stations, date, time, class, and optionally map selection.	
Description (Main Success Scenario)	Step	Action
	1	User enters departure station, arrival station, date, time, and optionally class (FR 1, FR 2).
	2	User may select stations on the map via the Mapping System (FR 11).
	3	User clicks “Search”.
	4	System queries the Train Database and retrieves available connections.
	5	System applies delay predictions from the Prediction System (FR 5, OB 2).
	6	System displays a list of results with train ID, intermediate stops, timings, track, total duration, and services (FR 3, FR 4).
	7	User optionally sorts/filters results by price, trip time, delay probability, or crowding probability (FR 5, NFR 5).
Extension	8	System updates the visualisation accordingly.
	Step	Other action
	1a	If the user selects an invalid station (e.g., same departure and arrival), system shows an error and asks for correction.
	4a	<i>If the Train Database query times out (>2 seconds, NFR 6), system retries once and then shows a partial result with a warning.</i>
	5a	If the Prediction System is offline, system shows results without delay predictions and logs the issue.
	6a	If no trains match the criteria, system suggests alternative dates or nearby stations.
Technology & data Variations	7a	If the user applies a filter that returns zero results, system displays “No trains match your filters” and offers to reset filters.
	→ If accessed from a mobile device, the map interface (FR 11) uses touch-optimised controls. → Delay prediction values are shown as “low / medium / high” probability or as a percentage, depending on the Prediction System’s output. → For anonymous users, search history is not saved; for registered users, recent searches may be reused (FR 17).	